

Joshua Cortes

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EDUCATION

California Baptist University | Riverside, CA | Expected 2027

B.S. Computer Science, Specialization in Machine Learning and Artificial Intelligence

California State Polytechnic University, Pomona (Cal Poly Pomona) | Pomona, CA | Aug 2023 - Dec 2025

Coursework in Computer Information Systems

EXPERIENCE

Software Engineering Intern

Dec 2025 - Present

Ormco (Envista)

Pomona, CA

- Build and deploy production .NET WPF applications (C#, XAML, MVVM) used daily by manufacturing line operators worldwide
- Barmold AQL: Led development of a computer vision pipeline (Basler cameras, Pylon SDK) that finds molds on trays and drives a Universal Robots arm through a custom C# Modbus TCP layer, automating inspection of 200+ trays/hr
- Trained and integrated an OCR model that reads engravings on resin molds and checks tray condition (present/absent, full/empty), moving QC checks onto the line and cutting inspection time by 80%
- Paperless: Moved the full work-order process off paper using .NET, MSSQL, and AWS S3, removing hundreds of printed pages a day and cutting order processing time by over 50%
- Rewrote an unstable RFID read/write tool from scratch as a multithreaded app, speeding up tag reads/writes and stopping up to 5 crashes a day under heavy load
- Shipped a cassette damage tracker and label generator (backed by AWS S3) to production at the Mexicali plant

IT Technician

May 2025 - Dec 2025

Alfa Business

Riverside, CA

- Provided Tier 1 to 3 support for hardware, networking, and software for 200+ users on Windows and macOS
- Managed Office 365 and Azure AD for 10+ companies, handling user setup plus onboarding and offboarding
- Wrote Bash, PowerShell, and Java scripts to automate antivirus detection, system and SQL Server monitoring, and firewall setup

PROJECTS

Bird Identifier | *Python, PyTorch, ONNX, FastAPI, React, TypeScript, Docker*

[Live Demo](#)

- Fine-tuned a ConvNeXt-V2 model to classify 555 North American bird species, reaching 89% top-1 and 98.8% top-5 accuracy on a 24,000 image test set
- Converted the model to ONNX for 1.4x faster CPU inference and shipped it as a live web app with a FastAPI backend, React/TypeScript frontend, and Docker

Stock Movement Predictor | *Python, scikit-learn, Flask, PostgreSQL, React, Docker, AWS*

[Live Demo](#)

- Built an ML pipeline over 1.27M rows of S&P 500 market data with time series cross-validation. Found and fixed a data leakage bug that was inflating accuracy, then added regression tests so it stays fixed
- Built a Flask REST API on PostgreSQL that serves live predictions, with password hashing, rate limiting, and SQL injection protection, deployed as one Docker image to Render and AWS Lambda

Book App | *React, TypeScript, Tailwind CSS, SQL, ESV API*

[Live Demo](#)

- Built a full-stack Bible app with login, verse highlighting, notes, and daily streaks, using the ESV API and shipping to desktop, iOS, and Android from one codebase

Asteroids Game | *Java, Swing, AWT*

[Live Demo](#)

- Built a multiplayer space shooter with three game modes and a 9-thread engine handling collisions, physics, player input, and enemy AI

TECHNICAL SKILLS

Languages: Python, C#, Java, JavaScript, TypeScript, SQL, Bash, PowerShell, HTML/CSS, XAML

Frameworks & Libraries: .NET, WPF, MAUI, React, PyTorch, ONNX, scikit-learn, Pandas, NumPy, FastAPI, Flask, Tailwind CSS, Swing/AWT

Cloud, Data & Tools: AWS (S3, Lambda), Docker, Git, PostgreSQL, Microsoft SQL Server, REST APIs, Azure AD, Office 365, Basler Pylon SDK, Modbus TCP, Google Colab, Atlassian

Concepts: Machine Learning (ML), Deep Learning, Computer Vision (CV), Optical Character Recognition (OCR), MVVM, Object-Oriented Programming (OOP), Data Structures & Algorithms, Multithreading